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A review of artificial intelligence methodologies in computational automated generation of high performance floorplans

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AI-driven energy-efficient floorplan generation is a transformative tool addressing energy challenges in building design driven by climate change, carbon emissions, and rising energy demands. This review explores Automated Floorplan Generation with Energy Efficiency Optimization (AFG-EEO) methodologies, integrating design generation, performance evaluation, and optimization to streamline architectural processes. It highlights gaps in methods, emphasizing user-centric enhancements, architect involvement, and future directions to improve methodologies, offering researchers, practitioners, and stakeholders valuable foresight.

Over the last decade, climate change has accelerated, causing a faster temperature increase than in the previous century^{1–9}. People spend more than 90% of their time indoors^{10,11}. There is a strong relationship between indoor environmental quality demand and carbon emission^{12–17}. On the other hand, the building sector consumes 40% of global energy consumption, which is expected to rise due to the impact of climate change and population growth^{1,18–22}. Moreover, the building sector is particularly concerned, responsible for a substantial 28% of total energy-related CO₂ emissions, mainly from electricity generation for building use^{7,12,23,24}. However, the design and construction of high-performance buildings have the potential to reduce energy consumption^{25–27} and Greenhouse gas emissions^{28,29} as reported by the International Energy Agency by 30% by 2050³⁰.

For this international trend, architectural design decision-makers should pay more attention to using energy simulation, artificial intelligence (AI)^{31–38}, and automated floorplan generation (AFG) tools to achieve high-performance buildings and net-zero energy consumption^{39–43}. Integrating AI into design processes represents a paradigm shift in the architectural field^{44–49}. This shift marks a fundamental change in architectural design's underlying principles, methods, and practices⁴⁵. Traditional sustainable design process typically begins with a detailed climate analysis⁵⁰, followed by multiple sketches to incorporate these findings, which is a time-consuming sequence^{51–54}. Subsequent steps involve manual iterations of floorplans and sections, tested through simulation tools, and repeated until a satisfactory design is achieved^{55–60}.

In contrast, AI tools have conducted a revolutionary approach that can rapidly generate comprehensive sets of floorplans that adeptly address climate concerns and functional configurations and yield high-quality renders

within minutes with a mere click or even a textual input devoid of specific climate analysis, sketches, or predefined design requirements⁶¹. In the meantime, it is essential to note that the integration of AI does not aim to replace architects with machine learning models; instead, it empowers them with a highly advanced foundation, eliminating the need to reinvent the wheel for each design. This paradigm shift represents a significant deviation from the traditional design process, offering architects a progressive starting point for their creative endeavors.

A lot of AI and AFG tools exist at architects' disposal. However, a critical observation shows that most of these tools are available in the market and prioritize factors such as maximizing floor space and facilitating fast design generation, typically catering to the real estate sector's needs.

One particularly promising application is using AI algorithm tools to generate energy-efficient floorplans. These tools still need to pay more attention to environmental considerations such as energy consumption, health, and well-being due to a lack of necessary parameters such as sun path diagrams, exterior temperature range, wind direction, and speeds. The floorplans may be spatially efficient but need environmental consciousness.

Several research prototypes have explored AI tools as promising candidates for architectural design despite their limited availability in the market (see Table 1). These prototypes emphasize generating functional floorplans and building masses^{62–66} that comprehensively account for factors such as energy consumption, daylight, Universal Thermal Comfort Index (UTCI), and carbon emissions, reflecting a comprehensive approach to architectural design that seeks to balance functionality and environmental responsibility. Currently, the market offers a variety of energy performance simulation tools, including CoveTool, OpenStudio, Honeybee, BEopt,

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Table 1 | A comparison of market-available tools versus advanced research prototypes (source: Authors)

Feature/Aspect	Market-Available Tools	Research Prototypes
Design Typology	Residential, schools, healthcare, mixed-use, etc.	Typically small residential floorplans
Generates	Generates both 2D and 3D models	Generally limited to 2D; A few consider 3D
Environmental Focus	- Treats environmental factors as complementary - Simple environmental analyses (sunlight, radiation, etc.)	- Strong emphasis on environmental factors - Integrated environmental considerations to shape the design (EUI, TC, PMV, SDA, ASE, etc.)
Design Process Inclusion	Environmental factors included at final stage	Incorporated throughout the design process
Hardware/Software	Robust capabilities	Limited hardware and software capabilities
User Interface	User-friendly interfaces, extensive docs	More complex, requires technical expertise
Performance	Rapid design iterations, and real-time feedback (significant time-savings)	Slower due to computational demands (Time-consuming)
Industry Integration	Seamless integration with software (Revit, Rhino, AutoCAD)	Less compatible, confined to specific platforms (e.g., Rhino, Grasshopper)

eQuest, and EnergyPlus^{67–69}. However, it is important to note that these tools primarily serve as evaluation instruments rather than generative tools. They require pre-existing architectural models for performance assessment instead of independently generating energy-efficient floorplans or building designs.

Both research prototypes and market-available automated floorplan generation tools have distinct advantages and limitations. Market tools can handle large-scale projects, such as schools, healthcare facilities, and mixed-use developments. They can produce both 2D floorplans and 3D masses with detailed layouts^{70,71}. In contrast, research prototypes are typically confined to small residential floorplans and often focus on 2D outputs, with limited forays into 3D modeling, which remain less advanced compared to market tools^{61,72–75}.

A key strength of research prototypes is their emphasis on environmental considerations, such as energy efficiency, carbon emissions, and thermal comfort, that are integrated throughout the design process^{61,76,77}. Conversely, market tools treat environmental factors as supplementary, often limiting their analyses to basic parameters like sunlight hours, radiation, and daylight^{78–81}. Such analyses may lack depth due to their reliance on less developed engines compared to tools like EnergyPlus and OpenStudio. Moreover, market tools generally introduce environmental assessments at the final evaluation stage rather than embedding them in the generative design process⁸².

However, research prototypes face notable constraints, including limited hardware and software capabilities compared to the robust functionality of market tools. Market tools are designed with user-friendly interfaces and workflow integration in mind, supported by extensive documentation and machine learning models, making them accessible to a broad user base. They are also optimized for speed, enabling the rapid generation of multiple design iterations and seamless integration into fast-paced workflows. In contrast, research prototypes are more complex and require significant technical expertise to run. Also, they are slower due to computational requirements and limited performance optimization⁸³.

Another advantage of market tools is their seamless integration with established industry software such as Revit, Rhino, and AutoCAD, facilitating efficient data exchange and visualization. Research prototypes, by comparison, are frequently restricted to specific platforms, such as Rhino and Grasshopper, which can limit their adaptability and compatibility with broader industry workflows.

The goal of integrating AI applications for the automated generation of energy-efficient floorplans is not to supplant the role of architects, which remains irreplaceable. Instead, the intent is to streamline the design process.

Traditionally, architects invest substantial time comprehending climate patterns^{84–86} and incorporating them into preliminary designs. Subsequently, they expend considerable effort using various tools to analyze energy consumption, daylight availability, carbon emissions, and material properties, iterating their designs based on each simulation's outcomes. This process is often repeated numerous times.

AI-driven automated floorplan generation serves as a time-saving catalyst by providing architects with energy-efficient preliminary schematic designs as a robust starting point^{87,88}. Architects can then optimize and refine these designs, applying specific simulation software such as special daylight autonomy (SDA) and annual sunlight exposure (ASE), also known as glare, with confidence, knowing that the generated floorplan represents the best-case scenario—effectively streamlining what would otherwise be a time-consuming and iterative traditional process.

The emergence of computational artificial intelligence (AI) in architecture

AI debuted in the architectural domain over five decades ago by Yona Friedman⁸⁹. Friedman's visionary work in the 1960s laid the cornerstone for infusing AI into architectural design processes. It spearheads the evolution of sophisticated AI-driven tools tailored for architectural applications.

Initially, Friedman emphasized conceptualizing computational algorithms oriented towards generating innovative architectural layouts, focusing on addressing urban challenges such as housing and urban planning. Concurrently, during this period, another significant development took shape as Peter Levin pioneered the automated generation of floorplans, introducing the concept of rectangular floorplans (RFPs) in 1964⁹⁰.

Researchers introduced graph-theoretical techniques in the 1970s and 80s^{91,92}. These techniques, aimed at constructing rectangular layouts, illustrate the correlation of functions within the floorplan using a relational graph and a tree structure. In 2012, a formula incorporating adjacency requirements was introduced, further advancing the field. Substantial progress continued to unfold in this domain until 2020, when a significant shift occurred. A prototype capable of generating dimensioned RFPs is introduced for any drawn rectangular arrangement with adjacency, size, and symmetric requirements during this juncture⁹³. This innovative approach considered crucial aspects like adjacency, size, and symmetry requirements, thereby elevating the sophistication of automated architectural layout generation.

Notably, while the predominant emphasis of these advancements lay in generating functional floorplans, the inaugural exploration into automated floorplan generation research focused on addressing environmental concerns emerged in 2013. Simulated annealing techniques are utilized to formulate meticulously optimized floorplan layouts, considering lighting, heating, and circulation parameters⁹⁴.

Before 2013, the research attempted to address energy-efficient floorplan layouts, which were perceived as optimization tools rather than comprehensive automated generation frameworks⁹⁵.

The workflow of generative floorplans algorithms

Generative design involves a multi-step process utilizing various algorithms tailored for each stage, including generators for creating new design studies, evaluators for assessing each study, and solvers for ranking and achieving design goals. These distinct algorithm types are categorized based on their

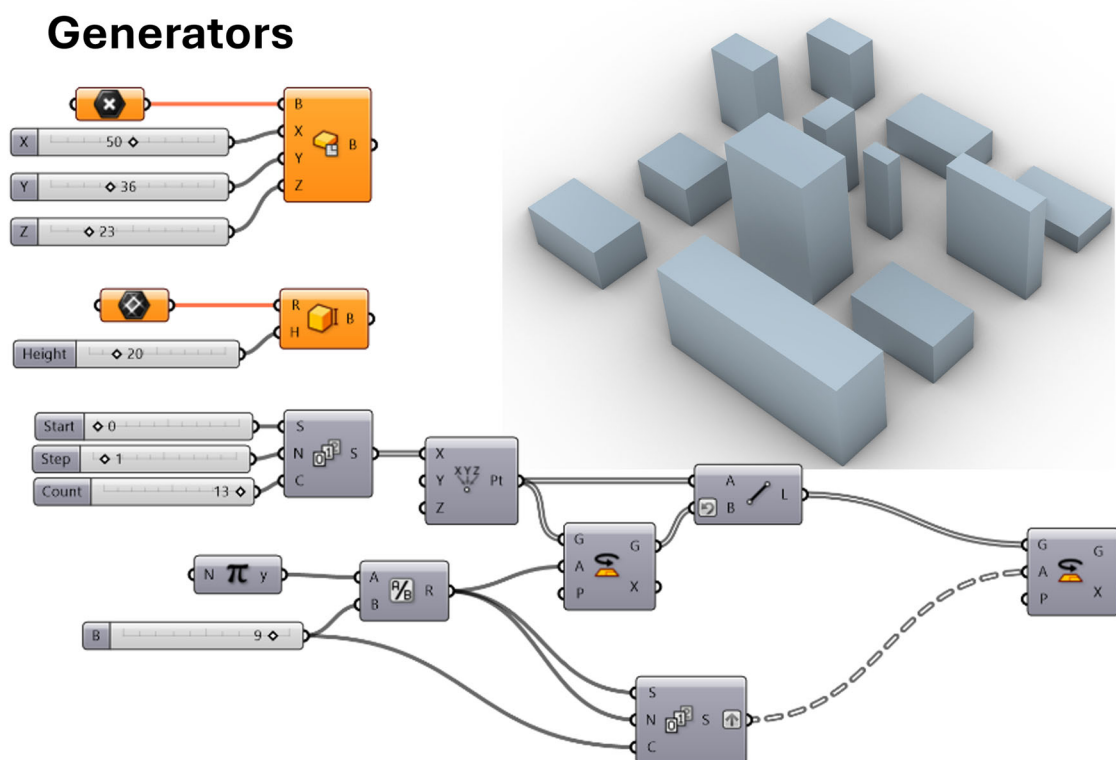


Fig. 1 | A Grasshopper script demonstrates the concept of generator algorithms responsible for creating several box design alternatives. (Source: Authors).

specific functions within the generative design process. This process presents thousands of design alternatives for architects that are narrowed down to give the best option based on the goals set in the algorithm.

Generators are the core of the algorithm workflow and are responsible for creating several design alternatives. They generate design configurations using simple or complex models to reach a model for evaluation and iteration (see Fig. 1).

Evaluators, the subsequent stage in the generative design workflow, face the intricate task of assessing the potential models created by generators. These evaluators can calculate the special daylight autonomy or the solar radiation falling on the building's façade during a specific day and time. Their work necessitates mathematical specifications, as they must provide a range of specific numbers to evaluate and compare design options (see Fig. 2).

Solvers, the last stage of the workflow, automatically rerun the script that contains both generators and evaluators several times to reach an ideal solution based on the data inputs as illustrated in Fig. 3. However, they require exact inputs to function accurately. Solvers employ various methods to run scripts; each offers a unique function. The current used solver methods include Randomize, which generates design options by assigning random values to input parameters for optimizing which conducts optimization runs by iteratively refining designs based on evaluator outputs; Cross Product, enabling exploration of the entire design space by combining parameter steps; and Like This, which introduces slight variations to existing input configurations for exploring different design variations.

So, in the current design, evaluators only evaluate after the generation process. The target is to move some of the primary energy performance data early in the generation process so it can be accounted for starting from the early phases of the design process.

Current research on the automated generation of high-performance floorplans

The current automated floorplan generation (AFG) concept is primarily oriented toward rapidly producing many design alternatives within a

constrained timeframe. This process traditionally prioritizes objectives like maximizing floor space, increasing unit counts, and establishing code-compliant boundaries^{93,96}. Several tools are available for real estate developers and architects to use in the market (see Table 2).

However, contemporary advancements in architectural design have given rise to a more refined approach known as the automated generation of high-performance floorplans. This evolution of AFG introduces a pivotal analysis phase into the conventional process. Its core objective is to yield a functional design that goes beyond the basics and considers various environmental variables meticulously. These variables encompass building orientation, architectural geometry, window-to-wall ratio (WWR), shading devices, and numerous other factors critical to achieving a high-performance design. Unlike the conventional AFG, which primarily addresses spatial aspects, this innovative approach extends its scope to enhance the building's performance across various sustainability and energy-efficiency metrics.

The automated generation of the floorplan with energy efficiency optimization (AFG-EEO)

The prospect of automated floorplan generation with optimized energy efficiency holds significant promise and is a crucial avenue for future research. This approach combines automated floor plan generation (AFG) with energy efficiency optimization (EEO), offering the potential to generate various layout alternatives within reasonable timeframes. Simultaneously, it allows for a comparative analysis of the energy performance of these alternatives while seeking out the most optimal designs. Performative computational architecture typically has three key components: floorplan generation, efficiency evaluation, and optimization, as outlined in previous research^{40,97–99}. Correspondingly, AFG-EEO encompasses these three fundamental aspects.

The AFG-EEO tool aims to integrate generative and optimization processes within a unified framework, introducing early energy optimization in conjunction with the AFG stage. In the AFG-EEO framework, the AFG segment primarily focuses on form-finding, incorporating algorithmic

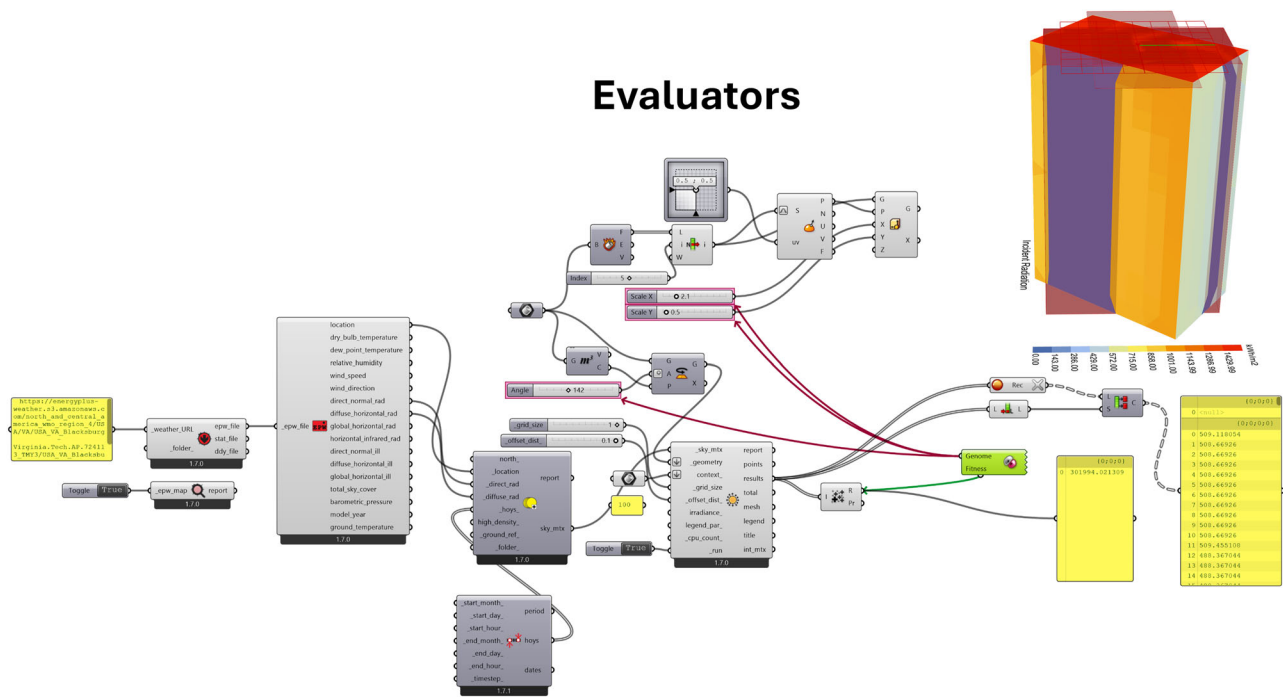
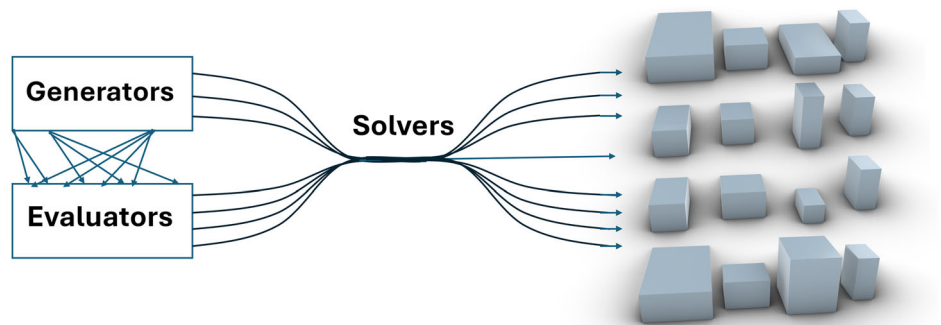


Fig. 2 | Evaluator algorithm in a grasshopper script for radiation analysis across multiple rotations and orientations. (Source: Authors).

Fig. 3 | An algorithm concept diagram illustrating the role of solvers in finding optimal solutions from generator and evaluator inputs. (Source: Authors).



design, associative geometry, and parametric design. Simultaneously, the efficiency assessment facet encompasses two critical elements: space functionality and energy performance, which are typically optimized or minimized as part of the overall objectives. The optimization component relies on specialized algorithms and encompasses the computational process of exploring various combinations of design variables. Its ultimate aim is to yield layout solutions demonstrating optimal values concerning the specified performance indicators. It is crucial to recognize that each of these segments possesses distinct requirements, and their interplay significantly influences one another when they are integrated. Therefore, considering their mutual impact, it is imperative to delve into the gaps and prerequisites for seamless integration.

Methodologies in current high-performance floorplan generation

The current methodologies for generating high-performance floorplans comprise three primary methodologies, which researchers adapt and tailor to varying degrees of complexity based on their specific objectives. These three methodologies are AFG, which focuses on floorplan functionality; EEO, which targets energy efficiency; and AFG-EEO, which integrates AFG and EEO. Researchers can delve deeper into these methodologies or streamline them, prioritizing one over the other

depending on their desired final output and goals. This adaptable approach allows a versatile, goal-oriented methodology for achieving high-performance floorplan designs.

AFG methodology

In the traditional automated floorplan generation (AFG) process (Fig. 4), the methodology begins with various data inputs, such as building typology, exterior floorplan boundaries, the number of rooms, and other relevant parameters. It uses spatial AI algorithms to represent floorplans, generating diverse floorplan sets. These generated floorplans are then evaluated based on predefined criteria. Particularly, the contemporary AFG approach incorporates a feedback loop that allows architects to assess the results and determine whether they meet requirements optimization. If necessary, the process can be repeated.

Real-world applications of floorplan generation in architectural practice

Many automated floorplan generation tools have become increasingly popular in architectural practice. Testfit, a tool that employs the AFG methodology, generates floorplans that integrate financial and technical data to maximize rental areas and profitability. Leading architecture firms such as Perkins & Will, WATG, DLR, and The Geyser Group have

Table 2 | A collection of contemporary automated floorplan generation software available in practice. (Source: Authors)

Software	Capabilities	Measures EUI	Used engine	Generates	Typologies	Scale	Firms using the software	Reference
Testfit	Floorplan configuration, conceptual design, financial data	No	Undetermined	2d & 3d	Multi-Family Single-Family Industrial Retail Hotel Parking	Urban and building scale	Perkins & Will WATG DLR Group BERARDI+ Studio hillier RBA Architects	128
SWAPP	Floorplan configuration, conceptual design, BIM Model	No	Undetermined	2d & 3d	Hospitality, Healthcare, Multifamily, industrials, education	Urban and building scale	Undetermined	70
Digital Blue Foam (DBF)	Floorplan configuration, operating costs, carbon emissions, solar radiation, sunlight hours, shading analysis, daylight autonomy, CFD analysis	No	Ladybug, TensorFlow, & Google Colab	2d & 3d	Small, medium & large-scale projects	Urban and building scale	TAKENAKA Jacobs AECOM Agency for Science, Technology and Research	78
Finch	Floorplan configuration, financial data, carbon footprint, energy, daylight, material, furnishing	No	Python, JavaScript TypeScript	2d & 3d	Residential, office, retail	Floorplan Scale	Herzog and de Meuron, White	79
Archi star	Floorplan configuration, code and permit compliance, Sunlight, ventilation, ground shadows	No	Undetermined	2d & 3d	All types of residential (boarding houses, townhouses, duplex, apartments, terrace houses), commercial, hotels, mixed use, parking	Building & urban scale	aurecon Government Architect New South Wales Hassell BUCHAN	80
Autodesk Forma	Design automations with Revit, operational energy, sun hours, solar energy, Optimize daylight potential, air flow, noise analysis,	yes	cloud computing	2d & 3d	All types	Building & urban scale	LandDesign CUBE3 Nordic Office of Architecture SWECCO	82
Delve	Massing, financial data, construction costs, configurations, solar energy, views, daylight access, sunlight hours, shadow hours	No	Undetermined	3d	Residential, office, retail, community, residential, parking	Building & urban scale	MODC BEYER BLINDER BELLE, QUINTAIN, Roche, City form Lab Heatherwick studio	129
Seismic School app	Floorplan configuration Sunlight hours Live traffic Flood data	No	Dynamo	2d & 3d	Schools	Building Scale	Limited to the United Kingdom	130
Prism app	Floorplan configuration, Weather data Comprehensive contextual data such as low emission zones, noise pollution, infrastructure, conservation areas, existing trees by type, size and height etc. Comprehensive actual 3d environment Flood data	No	Undetermined	2d & 3d	Housing and Mixed-use	Building and Urban Scale	Limited to London	71
Project refinery Beta (Primer)	Design support tool for Revit to, solar exposure, design possibilities Improve construction process, pattern configuration, and special data	no	Dynamo	2d & 3d	Supports the design of all buildings	Building & urban scale	Worldwide	81
Typenhaus+	Design configuration	No	Undetermined	2d & 3d	Housing	Building scale	Arnold and Gladisch Architects	131

Table 2 (continued) | A collection of contemporary automated floorplan generation software available in practice. (Source: Authors)

Software	Capabilities	Measures EUI	Used engine	Generates	Typologies	Scale	Firms using the software	Reference
Parametrische Planung	Design configuration, Views analysis Escape routes analysis Sun hours analysis	No	Undetermined	2d & 3d	Stadiums, opera houses, and theater buildings	Building scale	Undetermined	132
Share Form	Design support tool Solar radiation analysis, wind patterns, microclimate analysis, noise analysis	No	Undetermined	Analysis tool	Masses	Building and urban scale	Undetermined	133
Scout	Massing configuration, solar radiation, shading, views, outdoor comfort, Daylight, glare, walkability	Yes	Grasshopper	3d	Masses	Building and urban scale	Undetermined	134
Site solve	Design configuration, Views, sunlight hours	No	Undetermined	2d & 3d	Housing, parking	Building and urban scale	Undetermined	135
Omrt ostate	Design configuration, financial data, wind analysis, daylight, urban sound, sun & shading impact, views, Urban Heat Island, CO2	yes	Undetermined	2d & 3d	Undetermined	Building and urban scale	OZ Boom VASTINT SYNCHROON Heymans VAN ROEY PDR deKey ATENOR EDGE	136
Parafin	Design configuration, financial data	No	Undetermined	2d & 3d	Hospitality	Building and urban scale	Undetermined	137
Blox	Design configuration, financial data	No	Grasshopper	2d & 3d	All building Types	Building and urban scale	Gensler	138
PlanFinder	Single floorplan design configuration, Furnishing	No	Undetermined	2d	Residential, Commercial, Office	Floorplan Scale	Link Arkitektur Isthmus Sustainer B2Ai Architects ROOSROS	139

AFG methodology

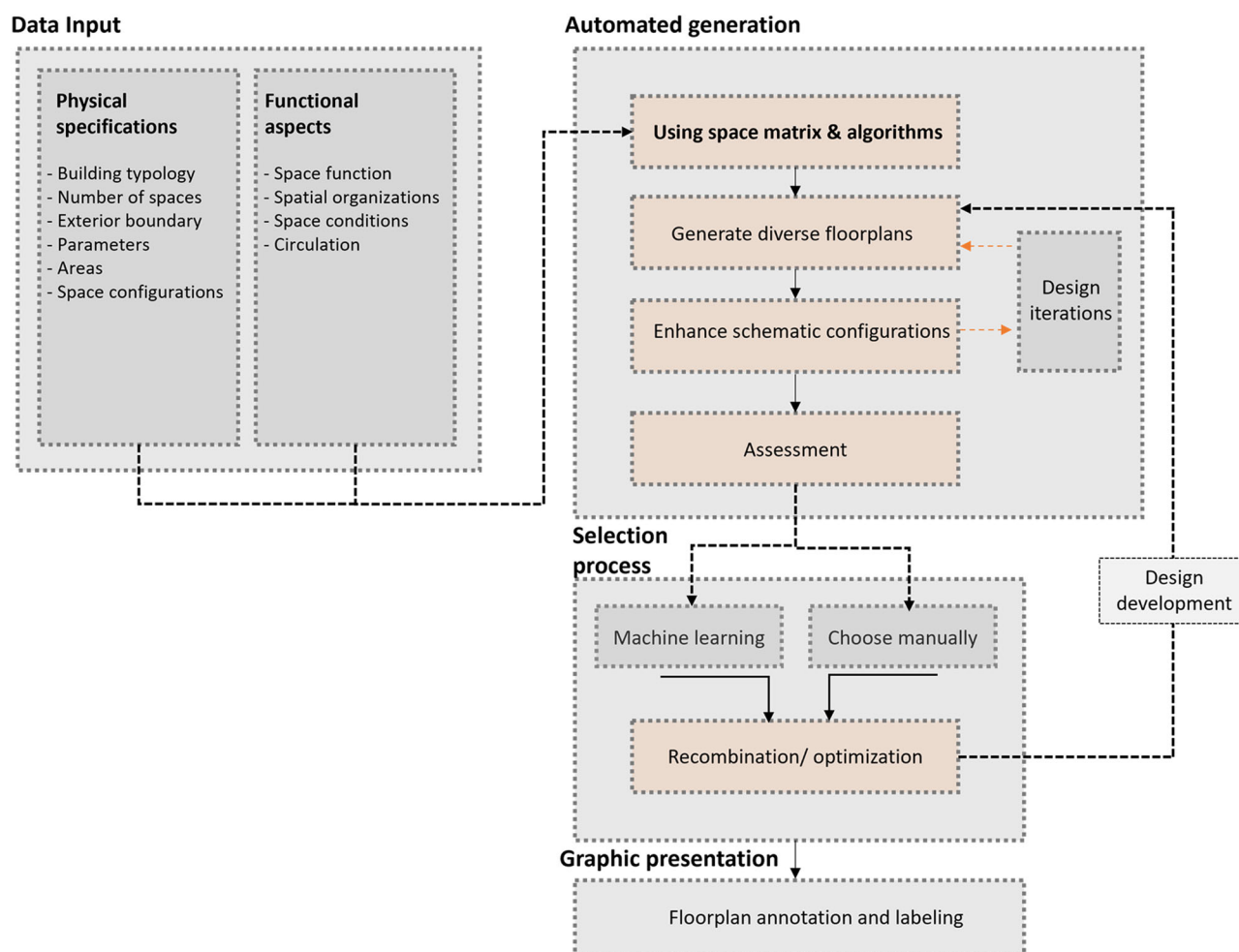


Fig. 4 | Automated floorplan generation (AFG) methodology. (Source: Authors).

incorporated Testfit into their design process. For example, The Geyser Group used Testfit in the ST. John mixed-use project in Austin, Texas. With Testfit, the team completed 40 design iterations in three months. Traditional methods would have allowed for less than 10 iterations in the same time-frame. The design team used different inputs in the tool to generate more design iterations that maximize space efficiency by increasing areas and the number of units (see Fig. 5). The project achieved a 13% increase in residential units, growing from 728 to 819, and expanded park space by 100%, increasing it from 1 acre to 2 acres¹⁰⁰. Similarly, the RESIA residential project demonstrated the efficiency of AFG tools. It significantly reduced site planning time and cut deal processing timelines from 14 days to 2 days. These improvements saved both time and costs significantly¹⁰¹. Similarly, Gensler employed gFloorz to design the Center for Global Health Innovation. The tool enabled the team to explore spatial configurations efficiently, though details about the process remain confidential¹⁰².

On the other hand, energy and Environmental Optimization (EEO) tools focus on sustainability. Tools like Autodesk Forma integrate environmental considerations such as energy use intensity (EUI), natural ventilation, and solar energy potential. These tools enable architects to achieve more sustainable designs. For instance, RSP Architects used Forma in the sustainability and resilience building project at the National University of Singapore. They adopted a co-design strategy that emphasized analysis over generative design. This approach increased on-site energy production by 6%¹⁰³.

EEO methodology

The energy efficiency optimization (EEO) methodology contains a systematic process (Fig. 6) that follows the AFG. It begins by selecting alternatives generated through the AFG. These alternatives have a meticulous process, including creating a 3D building model and an energy simulation model. The energy simulation is usually done through specific tools such as Design Explorer (Fig. 7), Honeybee, COVE tool and several other tools. These simulation tools use region-specific climate data and weather files. Therefore, outcomes may not be directly applicable elsewhere unless adapted to different contexts and regions.

During this simulation, the energy performance of the created model is assessed with predefined objectives and then presented. The process could be repeated several times if the results do not meet the desired performance objective to ensure a continuous cycle of refinement and optimization to achieve high-performance floorplan designs (Fig. 8).

Researchers have employed a workflow with a clear separation between the AFG and EEO methodologies^{73,86–88}. The AFG is comprehensively completed and optimized first, followed by the EEO process, resulting in two distinct levels of optimization—one for space functionality and another for energy efficiency. In contrast, a second workflow integrates this process by incorporating early energy optimization and evaluation during the initial floorplan generation within the early stages of the AFG process^{89,90}. This integrated methodology offers advantages regarding energy performance, as it establishes a feedback

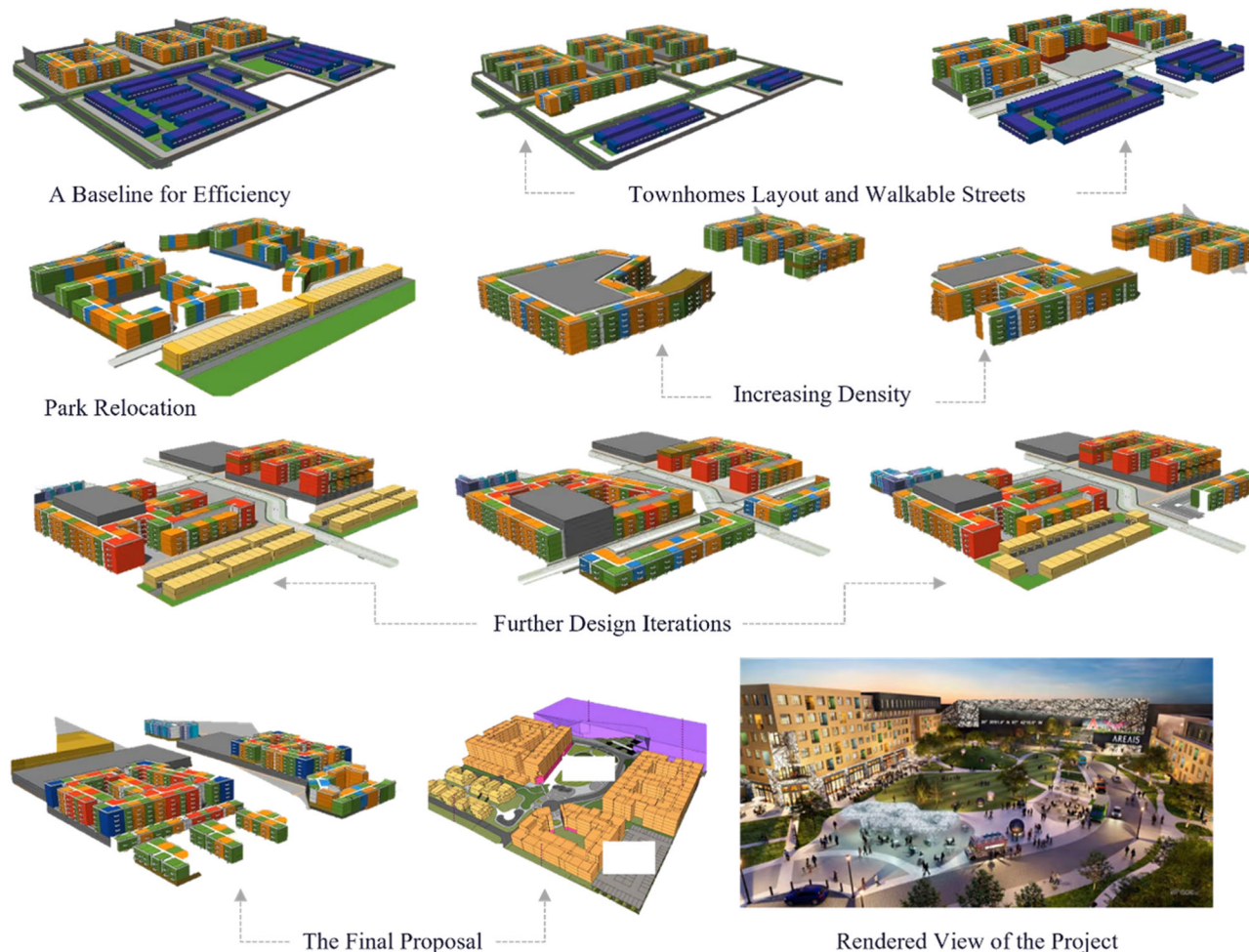


Fig. 5 | Design Iteration Process Using Automated Floorplan Generation (AFG) Methodology: A Case Study of the St. John Mixed-Use Development in Austin, Texas.¹⁰⁰

loop for AFG that considers functionality and energy efficiency from initiating the automated generation process.

AFG-EEO methodology

The AFG-EEO integrates AFG and EEO (Fig. 9). It has demonstrated the potential to significantly reduce cooling, heating, and lighting loads. To improve the AFG-EEO methodology, energy simulation engines such as EnergyPlus and OpenStudio should be incorporated into the simulation stage, as shown in (Fig. 9). This will ensure that energy efficiency is optimized from the early stages of the design process. EnergyPlus, developed by the U.S. Department of Energy, is a widely used energy modeling engine. It is often paired with complementary applications and plugins, including Honeybee and COVE Tool, which provide more specialized capabilities in energy and environmental simulations. EnergyPlus is part of the AFG-EEO framework, which plays a crucial role in analyzing the energy performance of generated designs. While EnergyPlus focuses on accurate performance analysis after creating a design, AFG-EEO generates energy-efficient designs in real-time. The key difference is that EnergyPlus provides analysis for completed designs, while AFG-EEO helps generate energy-efficient floorplans.

A conducted study generated a single floorplan to improve annual heating and cooling loads, predicted mean vote (PMV), and daylight levels using Ecotect as the energy simulation tool¹⁰⁴. The results revealed a remarkable 7.7% reduction in heating and cooling loads and 13% enhancement in PMV. Similarly, another study aimed to generate multiple floorplans that optimize annual heating and cooling demand using TECT with BIM. The results showed an 8.3% improvement in energy efficiency¹⁰⁵.

Other studies focused on thermal comfort by generating multiple floorplans using the Toolbox of Java integrated with EnergyPlus. These studies expressed a 29–33% enhancement in energy efficiency^{106,107}. However, to enhance the relationship between the process in terms of energy performance, a feedback loop between AFG and EEO should be applied. The energy data of the EEO will influence the floorplan optimization in the AFG of the AFG-EEO process. This process could identify weaknesses in the initial floorplan and generate optimized versions, establishing a continuous improvement loop until an optimal floorplan is reached. Therefore, AFG-EEO has a promising impact on the energy performance of building design. An integrated approach to how diverse climates, cultural factors, and building typologies affect AFG-EEO methodology would significantly improve the methodology's global applicability. Climate plays a crucial role in design performance, as different regions have varying requirements for thermal comfort, daylighting, and energy efficiency^{108,109}. For example, optimizing shading and minimizing solar heat gain in arid climates becomes essential^{110–113}, while maximizing natural ventilation and passive heating might be prioritized in temperate regions^{110,114}. Moreover, integrating cultural factors, including user preferences, local building traditions, and esthetic values, is significant for generating appropriate floorplans regarding energy efficiency and occupant satisfaction. For instance, incorporating traditional courtyard designs in Mediterranean contexts, which provide privacy and social spaces, exemplifies how cultural sensitivity can be integrated into the design process^{115–117}. Additionally, the building typologies, whether residential, commercial, or institutional, affect the applicability of specific AFG-EEO methods, as different types of buildings have unique usage patterns, occupancy schedules, and energy demands. By addressing

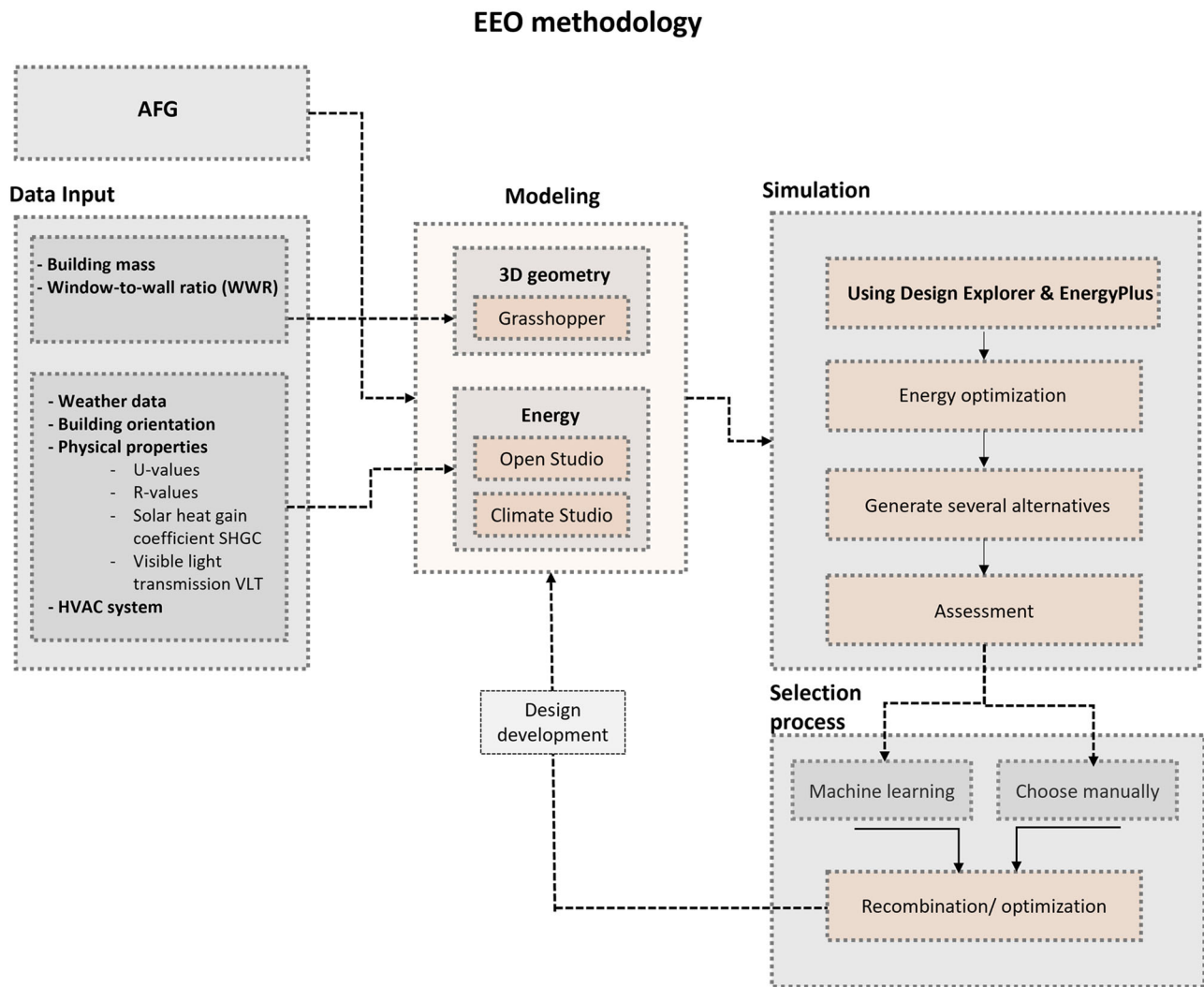


Fig. 6 | Energy efficiency optimization (EEO) methodology. (Source: Authors).

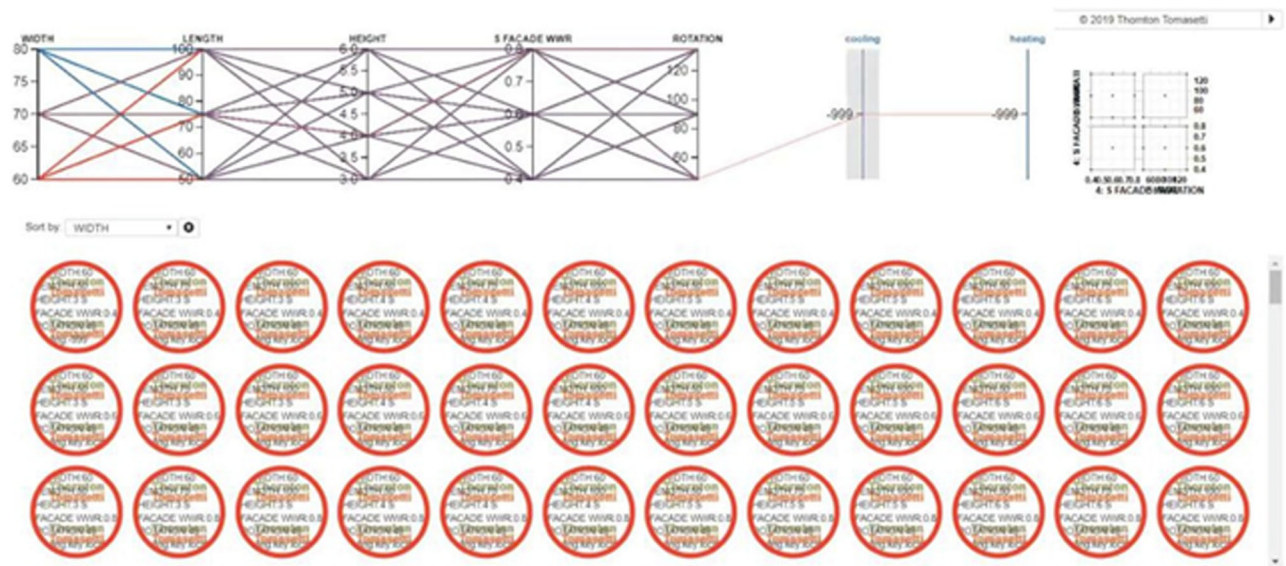


Fig. 7 | Energy simulation using Design Explorer. (Source: Authors).

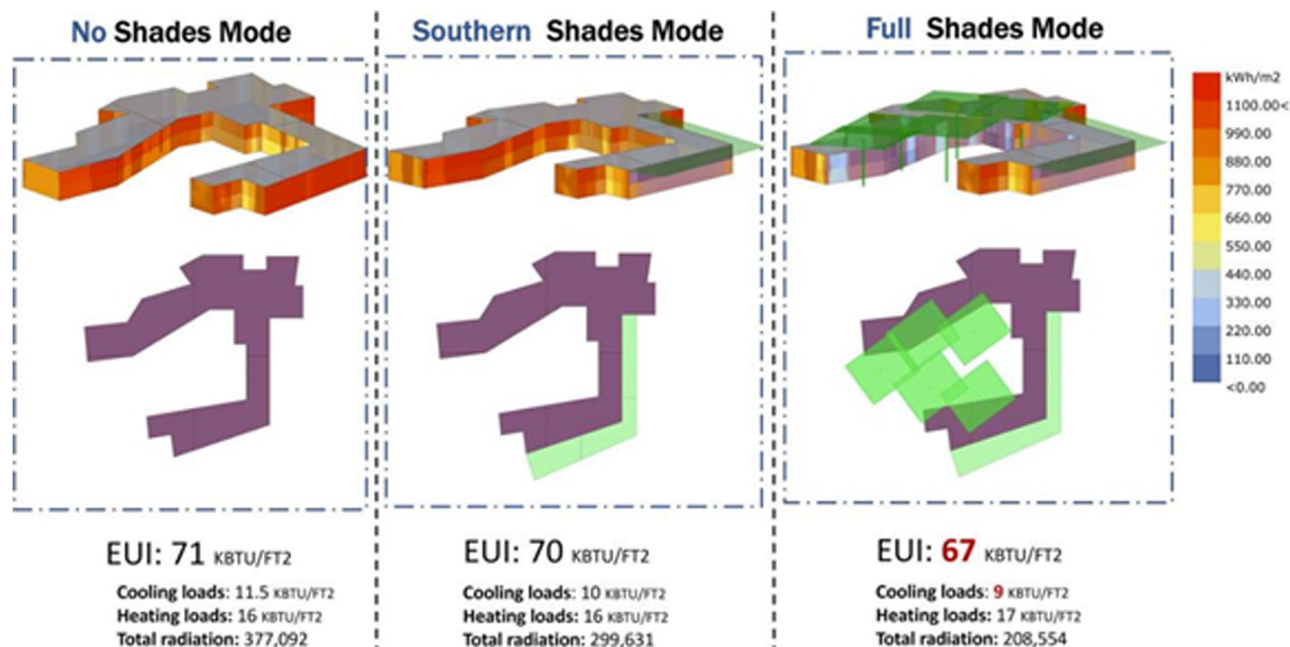


Fig. 8 | An example of energy optimization and assessment. (Source: Authors).

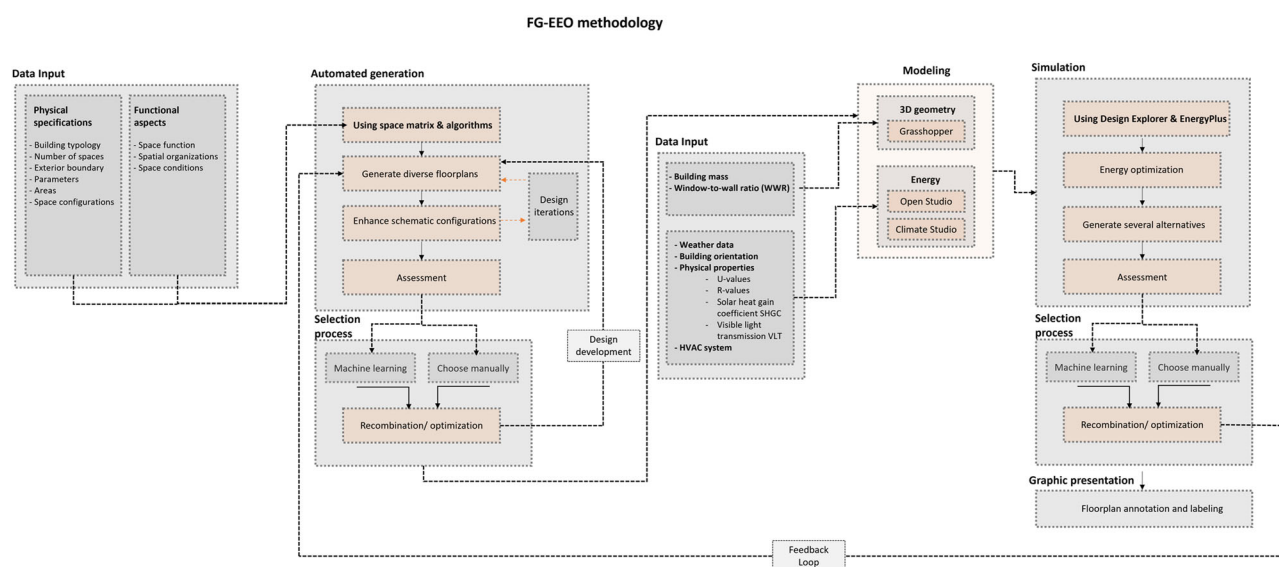


Fig. 9 | AFG-EEO methodology, which integrates AFG and EEO. (Source: Authors).

these factors, the AFG-EEO methodology can present a more holistic view and diverse global contexts, promoting sustainability and cultural relevance in building design.

Results

The previous sections show that the primary objective of the AFG-EEO methodology is to achieve a high level of energy efficiency accuracy for a wide range of generated floorplans from the AFG process within a condensed timeframe (see Fig. 9).

A methodology enhancement proposed by research begins with creating a parametric model focused on energy efficiency, then addressing the functional stage, and concludes with a second manual or automated EEO process⁹⁹. This distinctive methodology places the energy stage ahead of the functional stage, emphasizing energy considerations at the beginning.

While this approach may prove advantageous for architects knowledgeable in high-performance building principles, its applicability is

confined to a specific model, lacking diversity in the automated generation of floorplans. However, this approach enables the architects and students to establish correct orientations, geometries, window-to-wall ratios, and more. Conversely, it may present challenges for other architects and architectural students who are less experienced in sustainable design strategies, potentially yielding less favorable results.

According to the computational timeframe enhancement, energy simulations in architectural design are time-consuming, each taking minutes to hours. These simulations can become highly time-intensive when integrated into a comprehensive automated generation process. Some research employs the offline simulation method, which involves conducting all necessary simulations before optimization. This method allows rooms with similar conditions to share simulation results, significantly reducing the need to rerun simulations for each design solution during optimization¹¹⁸. Similarly, other research utilizes replacement models to emulate detailed simulation processes and predict outcomes based on

building information¹¹⁹ to reduce computational time compared to traditional simulation-based approaches.

Both methods offer practical ways to expedite energy simulations and enhance computational timeframes in architectural design.

Additional methods have been employed to simplify the complexity of architectural design models, simulations, and optimization processes¹²⁰. They adopted a sequential assessment approach involving a series of progression from more straightforward to more complex energy performance assessment techniques during optimization phases. This strategy, illustrated by a study employing simple yearly calculations, linearized convection calculations, and dynamic simulations, resulted in over two days of time savings compared to relying solely on complex simulations¹²⁰.

Another effective strategy is the hierarchical structuring of design variables¹²¹. This method organizes interdependent design variables into layers, preventing infeasible solutions and streamlining the generation process. Therefore, the hierarchical structuring of optimization objectives arranges objectives based on their impact on energy performance, enabling more efficient optimization efforts and reducing computational time.

Limitations of AFG-EEO

The AFG-EEO methodology is primarily applicable to small-scale projects, such as residential and single-story floorplans. However, applicability to large-scale projects, such as mixed-use, schools, or hospitals, has not been investigated due to the required time for simulation and complex spatial organization between many spaces. Also, it relies on specific generation and energy simulation tools, such as Design Explorer, EnergyPlus, and Climate Studio, which could restrict the generalizability of the findings to other contexts and regions. The methodology does not address qualitative design aspects, such as access to quality views and melanopic flux, which are important for enhancing circadian rhythms, health, and well-being. In addition, the AFG-EEO methodology requires powerful hardware and software, and it is time-consuming. The feedback loop between the generation and simulation phases needs to be repeated several times to achieve the desired results. It also focuses on quantitative factors, such as energy performance and space efficiency, while overlooking qualitative aspects, such as design esthetics and user experience.

Discussion

In this paper, contemporary methodologies for automated floorplan generation are reviewed and discussed in terms of energy cost minimization. For example, current AFG-EEO methodologies treat natural daylight as a variable in the energy equation, assessing its illuminance level that is primarily related to energy costs. However, it is essential to recognize that with significant advancements in LED artificial lighting, AI models may inadvertently consider replacing natural daylight with artificial lighting to achieve energy efficiency without realizing the potentially profound negative impacts on users' psychological well-being, mental health, and overall comfort. Research highlights that patients undergoing spinal surgeries in rooms with natural daylight views experienced reduced pain^{122–126}. The results illustrate a 22% decrease in analgesic medication, leading to a 21% reduction in medication costs and hastening recovery⁷⁴. With these findings and the advancements in artificial lighting, there is a compelling opportunity to reassess the integration of human-centric and experiential factors as essential data inputs in automated floorplan generation methods.

Although AFG-EEO methodologies prioritize 2D floorplan generation, they focus on moving environmental fundamentals to the early stages of the sustainable design process, such as geometry generation, building orientation, and facade design. These fundamental determinants can profoundly affect energy performance and warrant integration into the early stages of the design process to maximize their influence.

Furthermore, contemporary methodologies tend to curtail architects' roles, often casting them into an evaluative role during the latter phases of floorplan generation. In contrast, involving architects from the outset enables proactive control and design input and fosters a more collaborative and creative approach to energy-efficient building design. This approach

emphasizes the compelling need for a holistic and human-centered approach in developing future automated floorplan generation methodologies, highlighting the crucial role of architects in shaping the future of sustainable design.

Conclusions

AI-driven automated floorplan generation tools for energy-efficient buildings are evolving from functional-based designs to high-performance floorplans. AFG-EEO methodology offers automated floorplan generation, optimizing energy efficiency and balancing functionality and efficiency. Different workflows within AFG-EEO have shown significant energy consumption reductions in cooling, heating, and lighting loads.

Although advanced AFG-EEO methodologies, such as the energy-early approach, demonstrate significant potential, they require further refinement to support architects across varying levels of experience effectively. Strategies to streamline computational timeframes can expedite design processes. Nevertheless, gaps exist in current methodologies. Some AI tools prioritize energy cost savings at the expense of user-centric design considerations, such as the impact of natural daylight on well-being.

The integration of cross-disciplinary collaboration between architects, AI researchers, and environmental scientists is crucial in advancing AFG-EEO methodologies, particularly in projects that merge environmental performance and technological innovation^{41,47}. For example, Process Integration Platforms (PIPs) offer interactive tools that aid teams in managing multidisciplinary design and analysis processes, improving collaboration across project teams as well as within firms or industries to foster innovation and process improvement. Also, the Filter Mediated Design and Reference-based Optimization Method (RBOM), Multi-Attribute Interaction Design (MAID), Open Geospatial Consortiums AECOO, and Reference-based Optimization Method (RBOM) support collaborative generation and analysis of designs, particularly in environmental design optimization^{47,127}. AI researchers developed generative design algorithms that optimize multiple parameters, using tools like Grasshopper and EnergyPlus to quickly generate design iterations and provide architects with data-driven insights into performance optimization. This collaboration between architectural creativity and computational precision improved the adaptability of designs to climatic conditions while maintaining design integrity^{39,41}. Meanwhile, environmental scientists contributed essential data on site-specific climatic conditions, validating AI-generated designs to ensure that environmental performance metrics were both actionable and aligned with sustainability goals⁴⁷.

Reordering the design process to focus on sustainable elements and giving architects a more active role can enhance AFG methodologies.

In conclusion, AI-driven automated floorplan generation holds the potential to revolutionize architectural design. Bridging the gaps by embracing user-centricity, sequencing sustainable design early, and involving architects' creativity can optimize AFG methodologies for high-performance, energy-efficient buildings.

Future research

To enhance the utility of AFG-EEO (Automated Floorplan Generation with Energy Efficiency Optimization) for practitioners, future research should prioritize integrating user-centric metrics such as natural daylight exposure and indoor air quality, recognizing their significant impact on occupant well-being and productivity. Algorithms should incorporate health-related data to ensure designs prioritize daylight penetration and views. Simultaneously, optimizing the sequencing of sustainable design features, such as building orientation, geometry, and facade design, early in the generation process is essential to maximize energy performance. Feedback loops should reinforce these parameters during iterative refinements. However, this could be a time-consuming process. Architects should be actively involved, supported by user-friendly interfaces and real-time simulation feedback that facilitate creative decision-making.

Furthermore, integrating advanced simulation engines like EnergyPlus or OpenStudio within AFG-EEO frameworks can bridge the gap between

generative and evaluative processes, with surrogate modeling minimizing computational demands. Balancing energy efficiency with human-centric considerations, such as psychological comfort and user preferences, is crucial for meaningful adoption. Lastly, transitioning AFG-EEO tools from academic prototypes to market-ready applications requires enhancing scalability across diverse project typologies, improving interoperability with industry-standard platforms, and ensuring broader accessibility for practitioners. These steps collectively aim to refine AFG-EEO methodologies and make them more impactful in real-world architectural practice.

Data availability

No datasets were generated or analyzed during the current study.

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Author contributions

A.M.: Conceptualization, analysis, literature review, and interpretation of AI-driven energy-efficient floorplan methodologies; writing original draft; writing review & editing. A.A.: Methodology development; literature review; input on integrating user-centric elements and cultural aspects; writing review & editing.

Competing interests

The authors declare no competing interests.

Additional information

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